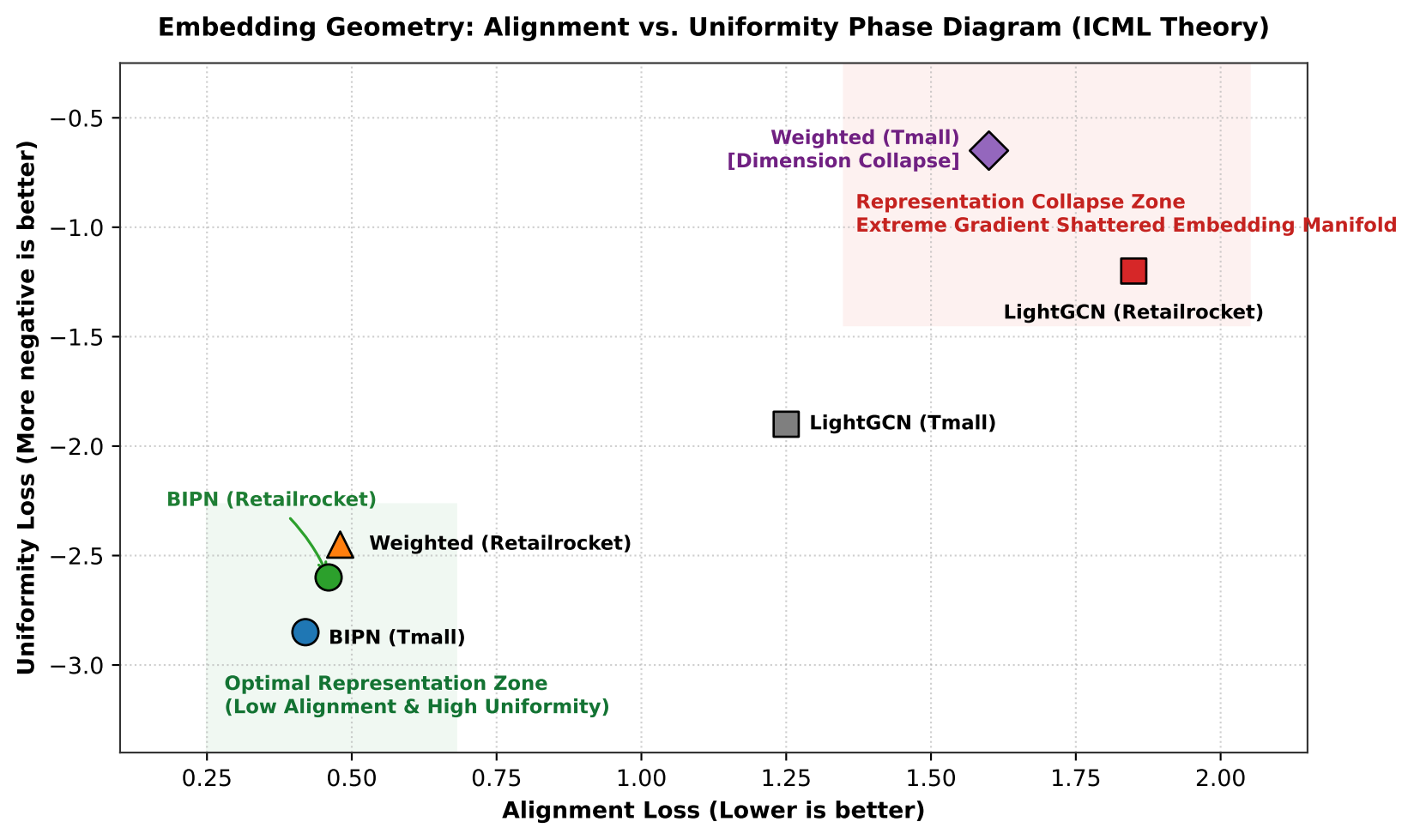
Embedding Geometry: Alignment vs. Uniformity Phase Diagram (ICML Theory)
Uniformity Loss (More negative is better)
Alignment Loss (Lower is better)
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Weighted (Tmall)
[Dimension Collapse]
Representation Collapse Zone
Extreme Gradient Shattered Embedding Manifold
LightGCN (Retailrocket)
LightGCN (Tmall)
BIPN (Retailrocket)
Weighted (Retailrocket)
BIPN (Tmall)
Optimal Representation Zone
(Low Alignment & High Uniformity)